

PROJECT REPORT

PROJECT NAME :

Design and Implementation of 9-Bit SAR ADC

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Objective

The objective of this project is to design and implement a 9-bit Successive Approximation Register (SAR) Analog-to-Digital Converter using Cadence Virtuoso. The project focuses on designing the comparator, SAR logic, and capacitive DAC at transistor level and verifying their operation through simulation.

1 Introduction

Analog-to-Digital Converters (ADCs) are essential in modern electronic systems because they convert analog signals into digital data. SAR ADCs are widely used because they provide good accuracy with low power consumption and moderate speed.

A SAR ADC uses a binary search algorithm to determine the digital output corresponding to the analog input signal. Due to its simple architecture and efficiency, SAR ADCs are commonly used in embedded systems, communication devices, and sensor interfaces.

2 Working Principle of SAR ADC

The SAR ADC mainly consists of:

- Comparator
- SAR Logic
- Digital-to-Analog Converter (DAC)

The conversion process works as follows:

1. The Most Significant Bit (MSB) is initially set to 1.
2. The DAC generates an analog voltage corresponding to the digital code.
3. The comparator compares DAC voltage with input voltage.
4. If input voltage is greater, the bit remains 1; otherwise it becomes 0.
5. The same process repeats for all remaining bits.

For a 9-bit ADC, the resolution is:

$$\text{Resolution} = \frac{V_{ref}}{2^9}$$

3 Architecture of 9-Bit SAR ADC

The complete architecture contains:

- Comparator
- Binary Weighted Capacitive DAC
- SAR Register Logic
- Clock Circuit

The comparator compares the analog input voltage with DAC output voltage. The SAR logic updates the digital bits sequentially, while the DAC generates analog voltages corresponding to digital codes.

4 Comparator Design

The comparator compares the analog input signal with the DAC output voltage. It produces a digital output depending on which signal is larger.

Important characteristics of comparator:

- High speed
- Low offset voltage
- Low power consumption

5 DAC Design

A binary-weighted capacitive DAC is used in this project. The DAC converts digital data generated by the SAR logic into corresponding analog voltages for comparison.

The DAC accuracy directly affects ADC performance.

6 SAR Logic

The SAR logic controls the conversion process. It updates the digital bits sequentially from MSB to LSB based on comparator output.

The SAR register stores the final digital output after completion of conversion.

7 Implementation

The complete design was implemented using Cadence Virtuoso at transistor level using CMOS technology.

Steps involved:

1. Comparator schematic design
2. DAC implementation
3. SAR logic design
4. Integration of all blocks
5. Simulation and verification

8 Simulation Results

The designed SAR ADC was simulated successfully. The comparator, SAR logic, and DAC operated correctly during the analog-to-digital conversion process.

Simulation results verified proper conversion of analog input signals into corresponding 9-bit digital output codes.

9 Advantages

- Low power consumption
- Good accuracy
- Simple architecture
- Moderate conversion speed

10 Applications

- Biomedical instruments
- Sensor interfaces
- Embedded systems
- Wireless communication systems

11 Conclusion

The 9-bit SAR ADC was successfully designed and implemented using Cadence Virtuoso. The comparator, SAR logic, and capacitive DAC were integrated successfully to perform analog-to-digital conversion.

Simulation results verified the proper operation of the ADC. The project provided good understanding of mixed-signal VLSI design and SAR ADC architecture.